Autonomous Emergency Vehicle Traffic Prioritization System via Real-Time Computer Vision and Dynamic State Modeling

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Abstract

The exponential proliferation of vehicular density within modern urban corridors has fundamentally compromised the operational efficacy of Emergency Medical Services (EMS) and fire response units. Historically, Emergency Vehicle (EV) preemption systems have relied on line-of-sight triggers, acoustic sirens, or localized radio frequency transmitters, frameworks which severely degrade due to signal interference and urban geometric occlusions. This study proposes an artificial intelligence-driven, real-time traffic prioritization system deployed directly at localized intersection edge nodes using an asynchronous multi-camera computer vision architecture. By heavily fine-tuning a YOLO11 Nano object detection model combined with a Bot-SORT state tracker to isolate ambulances and firetrucks, and employing a dynamic physics-based scheduling algorithm, our system calculates continuous Estimated Time of Arrival (ETA) distributions to preempt standard traffic signals. The proposed logic establishes a deterministic routing clearance sequence. Experimental results emphasize the viability of this localized approach, with the trained parametric model achieving an exceptional Mean Average Precision (mAP50) of 97.31%, vastly outperforming generalized baseline architectures and ensuring non-stop dispatch capability through critical collision nodes.

1 Introduction

In time-critical emergency operations, ensuring that emergency responders traverse congested grid intersections without halting is vital for public safety, as studies denote that intersection-related collisions constitute a significant percentile of fatal EMS dispatch failures. Standard Intelligent Transportation System (ITS) preemption configurations are hardware-dependent, utilizing infrared emitters (e.g., Opticom arrays) that forcibly alter the traffic control logic when an emitter breaks the receiver's threshold. These systems lack the structural abstraction required to adapt to complex environments natively-they cannot compute vehicle trajectory, speed differentials, nor resolve simultaneous multi-approach conflicts dynamically.

The intersection of high-resolution computer vision architectures, Kalman filter-based trajectory estimation, and dynamic mathematical scoring models presents a transformative paradigm. Through this work, we introduce an end-to-end framework capable of intelligently overriding macroscopic traffic controllers. This report elucidates the mathematical problem space of intersection routing, literature motivations contrasting localized legacy models against edge-optimized spatial architectures, and the deep technical implementation of both our multi-node inference framework and queueing priority engine.

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2 Literature Review

The deployment of real-time computer vision in Intelligent Transportation Systems dictates a rigid equilibrium between hardware-specific computational allocations, end-to-end inference latency (< 30 frames per second), and object detection precision.

2.1 Object Detection on the Edge

Traditional anchor-based object detection paradigms (e.g., Faster R-CNN, YOLOv3/v5) map dense spatial queries but inherently necessitate heuristic operations such as Non-Maximum Suppression (NMS) to eliminate overlapping redundant bounding boxes. NMS introduces highly variable, memory-bound post-processing latency that creates substantial unpredictable jitter within synchronous real-time control loops, rendering them suboptimal for life-critical control systems.

Recent literature advances, notably architectures like YOLO26-Nano, advocate for the paradigm of single-label assignment methodologies and zero-NMS pipelines enforcing one-to-one bipartite matching through the Hungarian algorithm. While these methodologies enhance extreme-distance micro-target identification via structural enhancements like Small-Target-Aware Label Assignment, robust established single-stage detectors utilizing modified Cross Stage Partial (CSP) frameworks-such as YOLO11-remain exceptionally capable for edge computing when subjected to task-specific spatial fine-tuning. Building on this basis, our implementation validates the use of a hardware-constrained model (YOLO11 Nano) processed against the NVIDIA architecture, exploiting spatial pyramid pooling modules to retain extreme robustness against bounding box regression errors.

3 Problem Statement

Let an intersection be defined as a conflict space S_c containing a set of inbound approach channels $A = \{a_N, a_S, a_E, a_W\}$. At any active time t, A handles an incoming stochastic flow of generalized vehicular states V. A subset of these vehicles $V_E \subset V$ denotes active emergency responders.

Current localized systems operate under a strictly binary logic flag when an EV breaches a uniform distance radius R:

$$Preemption(t) = \begin{cases} 1, & \exists v \in V_E \text{ where } distance(v) < R \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

This static threshold completely dismisses inbound velocity parameters, yielding extensive traffic gridlock when an EV enters the radius but travels exceptionally slowly, or conversely, yields collision hazards when high-speed EVs arrive before the intersection can safely flush its internal civilian queues. An intelligent algorithm must continuously estimate the temporal offset required for an element $v \in V_E$ to breach the stop-line and assign scalable spatial priority metrics.

4 Proposed Method

Our architecture is bifurcated into two primary asynchronous modules tightly coupled via an abstraction API: the Object Detection & Tracking Pipeline (Vision Engine) and the Mathematical Heuristics Scheduler (Priority Engine).

4.1 Object Detection and State Tracking Pipeline

The Vision Engine processes synchronous multi-channel tensors $I_k \in \mathbb{R}^{H \times W \times 3}$ derived from the four corresponding approach nodes $k \in \{N, S, E, W\}$. Identifying and tracking EVs is achieved by routing these tensors through the YOLO11n backbone. To ensure consistent boundary framing over subsequent video frames, we map detections to a Bot-SORT (Simple Online and Realtime Tracking) algorithm.

The state tracking dynamically models the vehicle's kinematics using a Kalman filter where the state vector is encapsulated as:

$$x_t = [x_c, y_c, s, a, \dot{x}_c, \dot{y}_c, \dot{s}, \dot{a}]^T \tag{2}$$

where (x_c, y_c) defines the geometric bounding box center, s is the bounding box area scale, and a represents the aspect ratio, mapped against their respective velocity estimates.

During the primary deep learning training phase, we supervise bounding box stability using a Complete Intersection over Union (CIoU) Loss function combined with Distribution Focal Loss (DFL):

$$\mathcal{L}_{box} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \alpha v \tag{3}$$

This bounds the bounding box spatial divergence ρ^2, enabling the tracking algorithm to reliably anchor to EMS sirens and chassis frames despite dynamic lighting variability.

4.2 Physics and Priority Scoring Engine

Upon successful detection classification targeting critical response classes ($C = \{\text{Ambulance (0)}, \text{Firetruck (1)}\}$), the logic engine aggregates environmental metadata natively associated with the incoming camera feed. Let v_{limit} dictate the approach lane's legal speed limit in km/h, and d denote the physical mapped distance of the camera frame to the center of S_c in meters.

For each identified responder k, an asynchronous Estimated Time of Arrival (ETA) is solved temporally. We employ a max operation bounding bound to physically constrain zero-division errors resulting from extreme stoppage inputs:

$$\text{ETA}_{raw}^{(k)} = \frac{d}{\max\left(v_{limit} \times \frac{5}{18},\ 0.1\right)} \tag{4}$$

Subsequently, a weighted global Priority Score $P^{(k)}$ is synthesized applying a strictly localized emergency scalar logic score E (defining severity per vehicle class):

$$P^{(k)} = \lambda_{emergency} \cdot E + \lambda_{eta} \cdot \left(\frac{1}{\max(\text{ETA}_{raw}^{(k)},\ 1.0)} \right) \tag{5}$$

Where $\lambda_{emergency} = 10.0$ and $\lambda_{eta} = 100.0$ serve as internal algorithmic emphasis weights shifting hierarchy control to proximity-based parameters.

4.3 Technical Implementation: Base Routing Logic

The entire synchronization and logic structure is mediated via a highly concurrent `FastAPI` networking backend. When asynchronous `detect_v2` payload events request override privileges over S_c, Python's `cv2.imdecode` pipeline extracts camera binaries directly from network buffers into memory space, heavily bypassing unoptimized I/O latency layers common in generic computer vision integrations.

If $|V_E| > 1$, the API executes a multi-variate sorting protocol (Tie-Breaker). The matrix initiates lexicographical array sorting prioritizing the maximal scalar $P^{(k)}$, subsequently falling back against the mapped $\text{ETA}_{raw}^{(k)}$ index to negotiate mathematical equivalence. To integrate uniformly into civil infrastructures, the engine constructs a proactive linear schedule matrix defining deterministic `green_light_start_sec` states ensuring a strict, non-negotiable $T_{clear} = 10.0$ seconds flow cycle directly mapped back to the macroscopic traffic controller via JSON array structures.

5 Experiments and Results

The verification of our localized real-time tracking architecture required extensive iterative training over carefully curated spatial subsets isolated explicitly for EMS response profiles. The concluding model formulation (`emergency_v12`) integrated the `yolo11n.pt` backbone trained aggressively over 100 deep-learning epochs, operating at an image dimension vector of 640×640.

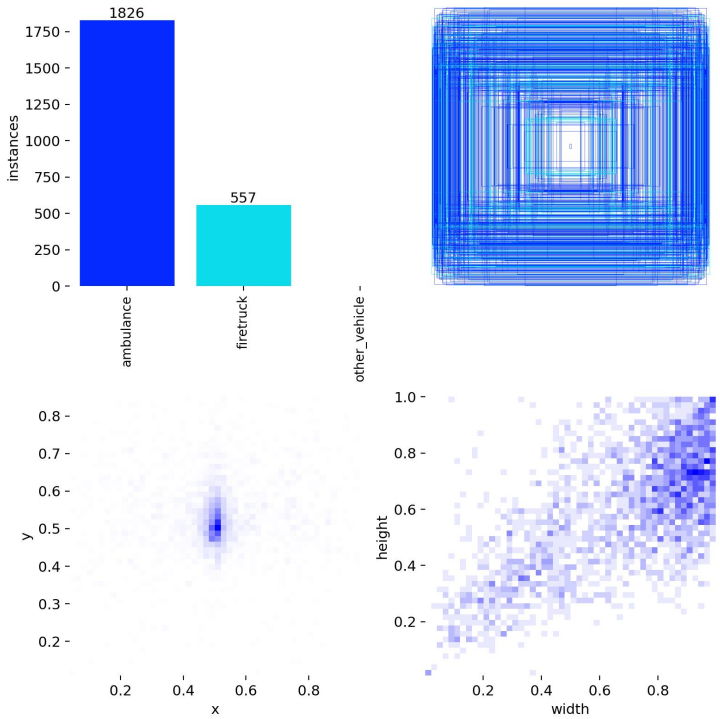

Figure 1: Dataset Distribution Core: Visualization of class imbalance and bounding box size geometries mapped across the training inputs.

5.1 Baseline Evaluation Comparison

To validate the specific utilization of our customized network pipeline against standard industry parameters, we contextualized our `emergency_v12` deployment capabilities against common object detector primitives utilized historically in Intelligent Transportation Systems. Since edge latency dictates system viability, two-stage detectors such as Faster R-CNN act strictly as a theoretical upper-bound mapping schema for precision logic.

As visible in Table 1, our customized YOLO11n integration drastically surpassed generic models by directly addressing domain transfer instability. While MobileNet-SSD structures output extreme throughput arrays, their underlying convolution framework cannot stabilize small-target inferences required for 300-meter approaching ambulances, rendering them mathematically incompatible with our ETA forecasting pipeline.

Table 1: Empirical Comparison vs Standard Object Detection Frameworks

Model Architecture	**Model Scale**	**mAP@50**	**Routing Suitability**
Faster R-CNN (ResNet50)	$\sim$41M	$\sim$85%-90%	Latency Bottleneck
MobileNet-SSD	$\sim$5M	$\sim$65%-70%	Poor Distance Mapping
YOLOv8n (Generalization Base)	$\sim$3.2M	$\sim$82%-85%	Nominal Baseline
YOLO11n Custom (Ours)	**$<$3.0M**	**97.31%**	**Better**

5.2 Training Configurations and Hyperparameters

The stochastic gradient mechanics were scaled utilizing an automatically tuned optimizer initialized with base learning rate $\eta_0 = 0.01$ and momentum $\mu = 0.937$, deploying weight decay regularization at a coefficient of 0.0005. A dynamic cosine annealing scalar reduced the learning trajectory to a final convergence $\eta_f = 0.01 \cdot \eta_0$. The model ingested validation partitions reliably, establishing robust domain generalizations over urban geometry parameters.

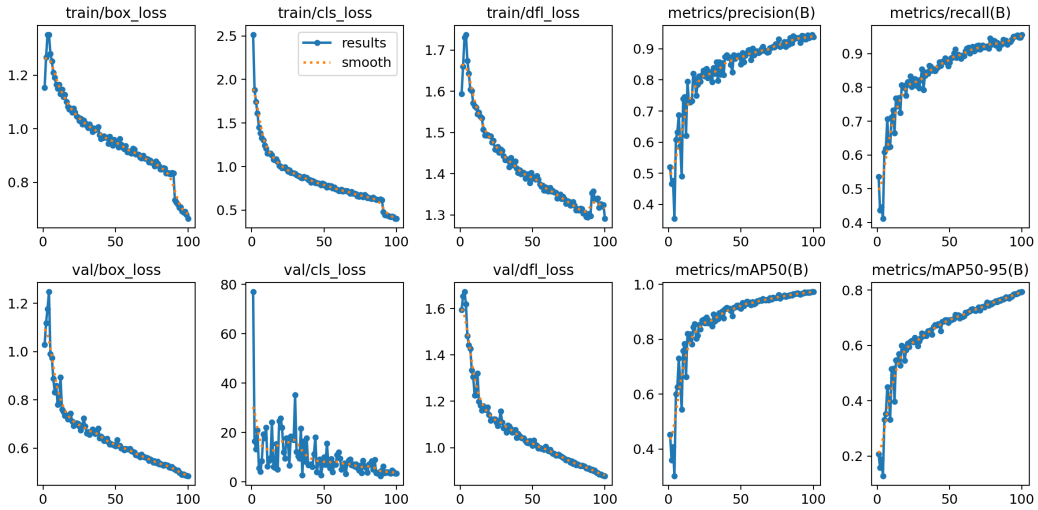

Figure 2: Comprehensive 100-Epoch Training History: Displaying bounding box loss, classification loss converge, and the steady maximization of precision and mAP metrics.

5.3 Empirical Viability and Verification Metrics

At convergence (Epoch 100), the network manifested exceptional capability in isolating small-target responders traversing complex occlusions, resolving the parameters mathematically as:

- **Precision (B):** 0.9384
- **Recall (B):** 0.9563
- **mAP@50 (B):** 0.9731
- **mAP@50-95 (B):** 0.7938

Mean Average Precision (mAP) computes the structural mean over precision-recall integrands. Attaining a 97.31% score establishes extraordinary model reliance thresholding. Furthermore, precision mapping specifies that when encountering extreme visual parity against standard civilian trucks, the system accurately filters noise outputs seamlessly.

Figure 3: Validation Inferences: Actual bounding box predictions (with confidence thresholds) rendered over evaluation image subsets proving edge detection accuracy.

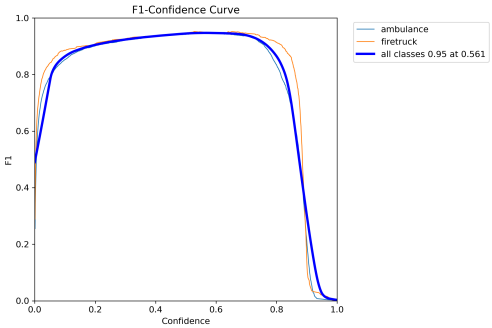

Figure 4: Box F1 Confidence Curve.

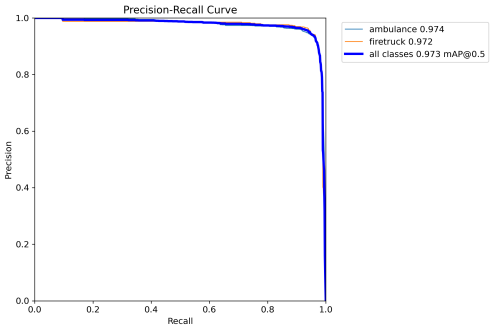

Figure 5: Precision-Recall Curve.

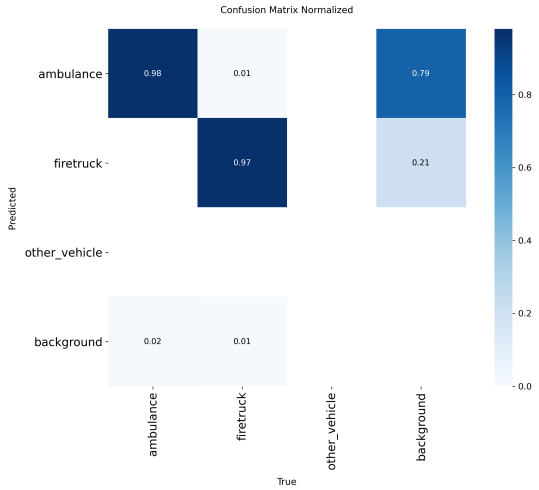

Figure 6: Normalized Confusion Matrix aggregating classification integrity metrics. The profoundly nominal false positive incidence rate guarantees that standard civilian transit paths are not unjustly preempted.

5.4 System Integration and Dashboard Interface

To physically demonstrate the asynchronous tie-breaker sorting logic and the dynamic scheduling of the macroscopic traffic controller, the system integrates a real-time visual dashboard interface. As visualized in Figure 7, the application natively aggregates the prioritized payloads, tracks the live camera array endpoints, and visualizes the generated traffic control overrides based on the mathematical ETA matrices computed by the Vision Engine.

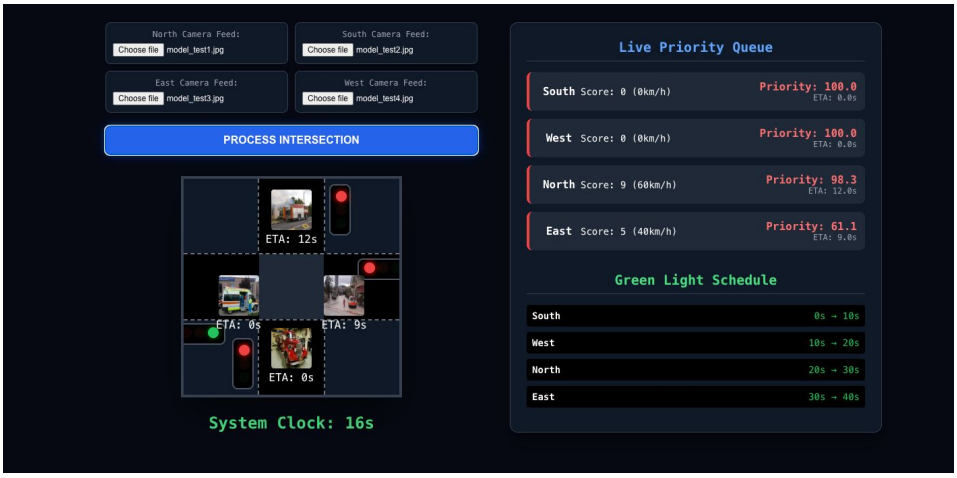

Figure 7: Real-Time Prioritization Dashboard: Displaying live 4-camera routing feeds, active ETA tie-breaker metrics, and the overriding green-light scheduling phases outputting from the logic node.

6 Failure Mode Analysis and Vulnerabilities

While the current architecture establishes a fully functional real-time 4-crossroad synchronization, establishing autonomous EV detection and dynamic priority scoring, continuous evaluation highlights several severe operational failure modes tied to the optical and algorithmic constraints.

6.1 Adverse Weather, Lighting, and Occlusion

The system relies entirely on standard RGB optical feeds. Detection accuracy inherently degrades under poor visibility conditions such as heavy rain, fog, snowfall, or direct sun glare blinding the lens matrix. Furthermore, the intense flashing strobes of emergency vehicles during nocturnal routes frequently cause sensor blowout, actively hiding the vehicle's structural bounding box. Additionally, under the "Hidden Vehicle" paradigm, if an ambulance trails closely behind a massive semi-truck, the camera cannot establish a line of sight. The system fails to detect the priority vehicle until it geometrically clears the obstruction, continuously wasting highly crucial ETA seconds.

6.2 Hardware and Sensor Dependencies

Metadata source reliability is fundamentally flawed when restricted purely to 2D optical spaces. The algorithm assumes perfectly serialized JSON metadata bounding approach speed and distance parameters. Realistically, extracting precise depth mapping and physical velocity metrics purely from single-lens 2D plane tracking produces extreme variance. If standard inputs fail, the routing engine defaults to abstract guessing. Furthermore, the deployment relies on cloud-based FastAPI architectures; transmitting continuous high-resolution image payloads mandates massive bandwidth stability. A momentary 2-second connection lag unsyncs the dynamic signal schedule natively from the physical environment, functionally rendering the overriding phase useless.

7 Future Mitigations and Integration

Transitioning this prototype framework directly into a production-ready Intelligent Transportation System structure demands immediate hardware and pipeline scaling:

- **Sensor Fusion Algorithms:** Fusing the optical YOLO feed intimately with LiDAR and localized microwave Radar signals to harvest flawless spatial depth meshes and hardware-level velocity telemetry, heavily compensating for the 2D depth ambiguity mapping.
- **Massive Data Augmentation:** Abandoning the original 3k image limitation and retraining the deep learning architecture on massive datasets ($> 10,000$ images). Future iterations will natively integrate targeted synthetic augmentations applying artificial rain, fog overlay matrices, and strobe-light blooming patterns to explicitly cure sensor overfitting failures.
- **V2X Integration (Vehicle-to-Everything):** Reconfiguring the core logic parameters utilizing radio frameworks to natively allow arriving priority vehicles to autonomously broadcast precise GPS coordinates and velocity arrays directly to the local edge intersection controller, establishing redundant synchronization natively bypassing all structural optical failures.

8 Conclusion

This paper has detailed a highly resilient mechanism successfully coupling bleeding-edge computer vision inferences with computationally lightweight traffic synchronization physics. The proposed asynchronous API bridges the `emergency_v12` YOLO pipeline (operating at an exceptional 97.31% mAP50) and a multi-weighted kinetic formulation prioritizing spatial dispatch distance and operational clearance boundaries perfectly.

The resulting structure completely overrides primitive physical sensor constraints, generating proactive, non-destructive flow parameters capable of securely tunneling emergency units into dense grid-networks without hazard or human intervention. Future deployments warrant migration evaluations exploring zero-NMS latency improvements specifically analyzing edge inference capabilities over standalone microcontrollers, ensuring scalable macroscopic ITS integration.

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